

MTH203: Multivariate Calculus

Monsoon 2025

Name: _____

Roll No.: _____

Tutorial Quiz 10

Question:

1. Use the surface integral in Stokes' Theorem to calculate the circulation of the field

$$\mathbf{F} = (y^2 + z^2)\mathbf{i} + (x^2 + y^2)\mathbf{j} + (x^2 + y^2)\mathbf{k}$$

around the curve C : The square bounded by the lines $x = \pm 1$ and $y = \pm 1$ in the xy -plane, counterclockwise when viewed from above. (1)

2. Use the Divergence Theorem to find the outward flux of

$$\mathbf{F} = \sqrt{x^2 + y^2 + z^2}(x\mathbf{i} + y\mathbf{j} + z\mathbf{k})$$

across the boundary of the region D : the solid region $1 \leq x^2 + y^2 + z^2 \leq 2$. (2)

Please note: There is no fractional marking. Only integer marking.